Answer **one** question from this section in the spaces provided.

7 **(a)** A small positively charged particle X of mass 1.00 x10^{-6} kg and charge 1.60 × 10^{-12} C is travelling through a vacuum between two parallel electric plates.

It is moving horizontally at a constant velocity of 2.80 x 10^{5} m s^{-1}, as shown in Fig. 7.1. The two parallel electric plates are 5.00 mm apart and the top electric plate is held at -15.0 kV.

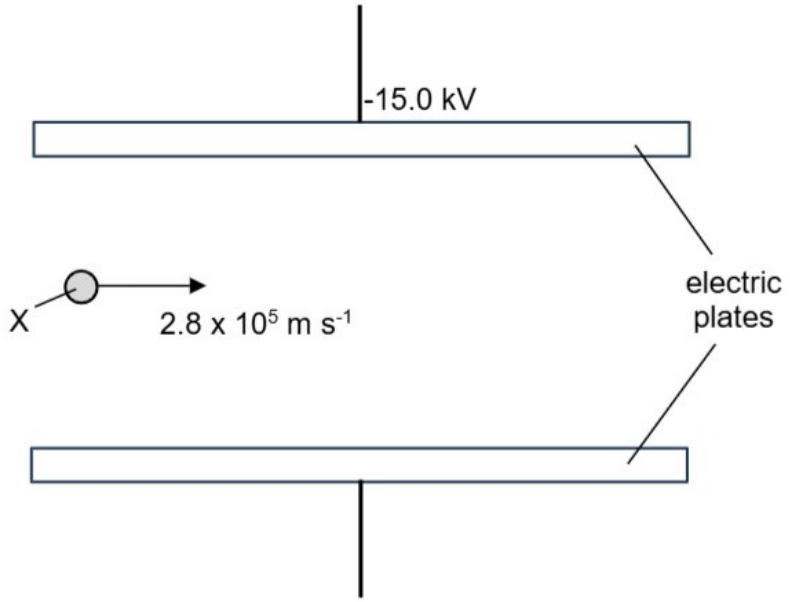


Fig. 7.1

(i) On Fig. 7.1, draw and label the forces acting on particle X. [1]

(ii) Define *electric field strength*.

.....

.....

..... [2]

(iii) Show that the electric field strength between the two electric plates is 6.1 x 10^{6} N C^{-1}.

[2]

(iv) Hence, determine the electric potential of the bottom electric plate V_{bottom}.

V_{bottom} = ... kV [2]

(v) State and explain what would happen to the path of particle X if

(A) the separation between the two electric plates is now increased.

.....

.....

.....

..... [2]

(B) the horizontal velocity of the particle is halved.

.....

.....

..... [1]

(b) Two charged metal spheres P and Q are placed 5.00 cm apart, isolated in space. The variation with distance x of the electric potential V between the spheres along the line joining their centres is shown in Fig. 7.2.

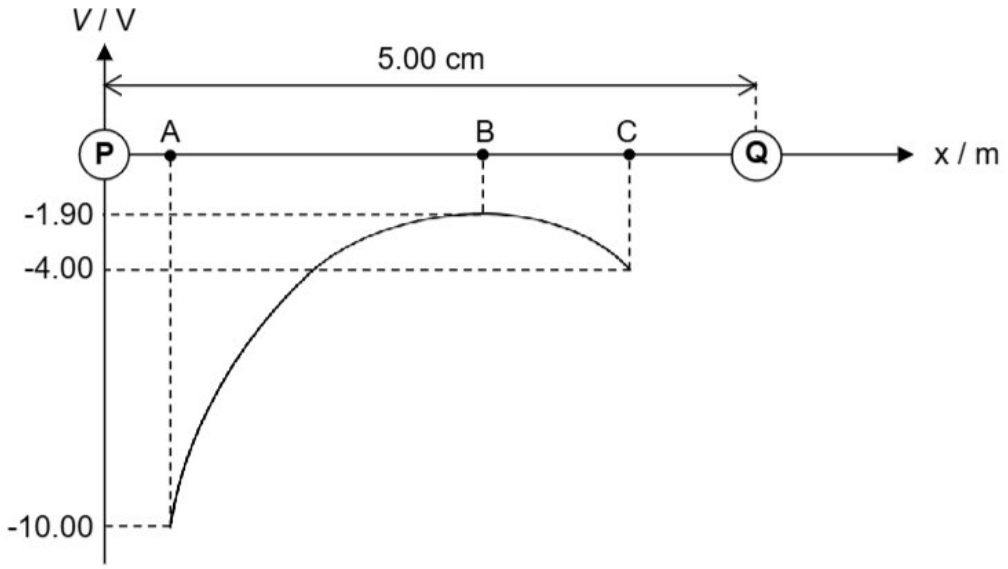


Fig. 7.2

At point B, which is 2.75 cm away from sphere P, the electric potential V is maximum.

(i) State the relationship between the electric field strength E and electric potential V.

.....

.....

..... [1]

(ii) On Fig. 7.3, sketch the variation with distance x of the electric field strength E between the surfaces of spheres P and Q. Numerical values are not expected.

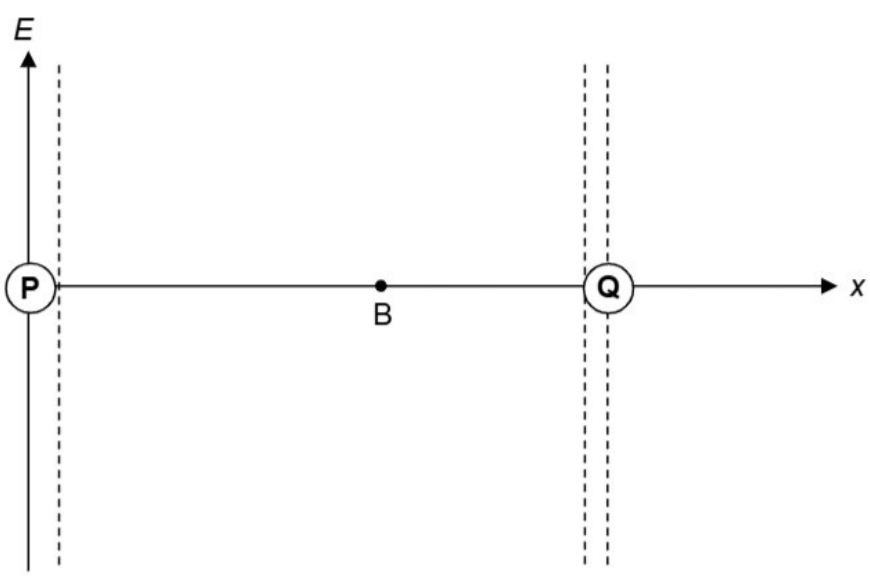


Fig. 7.3

[2]

(iii) Determine the work done by an external force in moving a positive charge of 1.20 nC from point A to point C.

work done = ... J [2]

(iv) State and explain the signs of the charges on spheres P and Q.

.....

.....

.....

..... [2]

(v) The ratio of the charge on sphere P to the charge on sphere Q (ignoring their signs) is 3:2.

Using Fig. 7.2, determine the charges on sphere P and sphere Q.

charge on sphere P = ... C

charge on sphere Q = ... C [3]

[Total: 20]